thws
Technical University of Applied Sciences Würzburg-Schweinfurt

KIWi – KI-Wissensassistenz

AI-Powered Knowledge Assistance for the THWS CampusPortal

Retrieval-Augmented Generation (RAG) | Local LLM | University Knowledge Retrieval

Problem

- University information is distributed across different sources (e.g., CampusPortal, wikis and manuals).
- Finding relevant and up-to-date answers is time-consuming.
- Traditional search can struggle with natural-language questions and process-related context.
- University information requires privacy-conscious processing.

Objectives

- Enable **natural-language question answering** about university processes.
- Retrieve relevant information from **approved knowledge sources**.
- Generate context-aware answers using RAG.
- Provide answers with source references and URLs.
- Support a **locally hosted solution** without external LLM APIs.
- Evaluate retrieval quality, correctness and response time.

Methodology

Data Ingestion
Available university knowledge records are collected from CSV files, combining synthetic data and publicly available THWS information.

Data Preprocessing
Questions, answers, key facts, evidence and source details are cleaned and prepared for further processing.

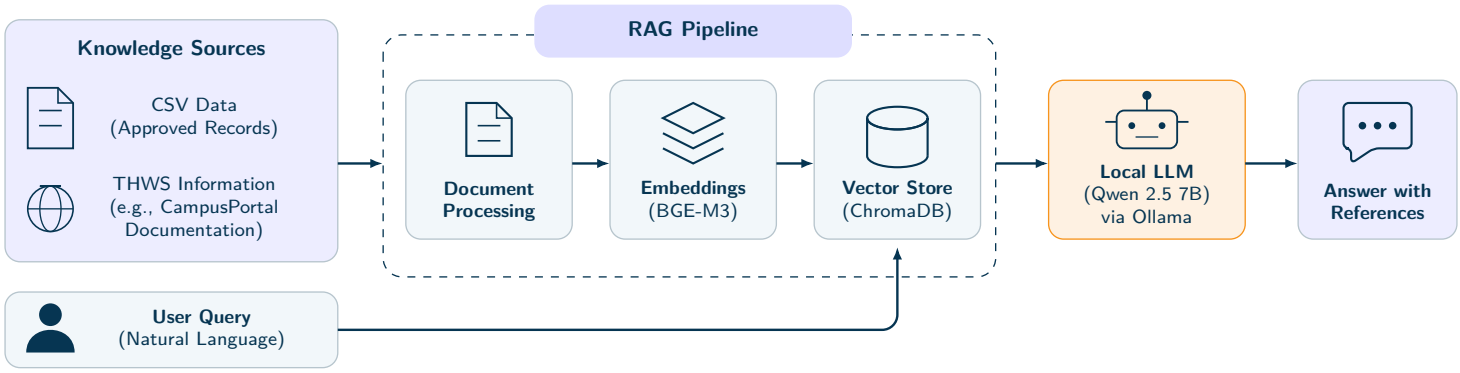
Text Chunking
Longer records are divided into smaller text passages while preserving relevant metadata and source information.

Embedding Generation
BGE-M3 converts text passages into numerical vector representations for semantic search.

Information Retrieval
ChromaDB stores embeddings and retrieves relevant passages based on the user's natural-language query.

Answer Generation
Qwen 2.5 7B Instruct, running locally through Ollama, generates context-aware answers using retrieved information and provides supporting source references.

Architecture



Deliverables / Expected Outcomes

Deliverables

- Working local **RAG prototype**.
- Automated CSV knowledge-ingestion pipeline.
- Persistent ChromaDB vector database.
- Terminal-based question-answering system.
- Source-supported answer generation.

Expected Outcomes

- Faster access to university information.
- More transparent, evidence-grounded answers.
- Reduced reliance on external AI services.
- Foundation for future CampusPortal integration and a web-based interface.

Team Members
Member 1 | Member 2 | Member 3

Supervisor
Prof. Dr. Ivan P. Yamshchikov

MAI Project
WS 2026/27